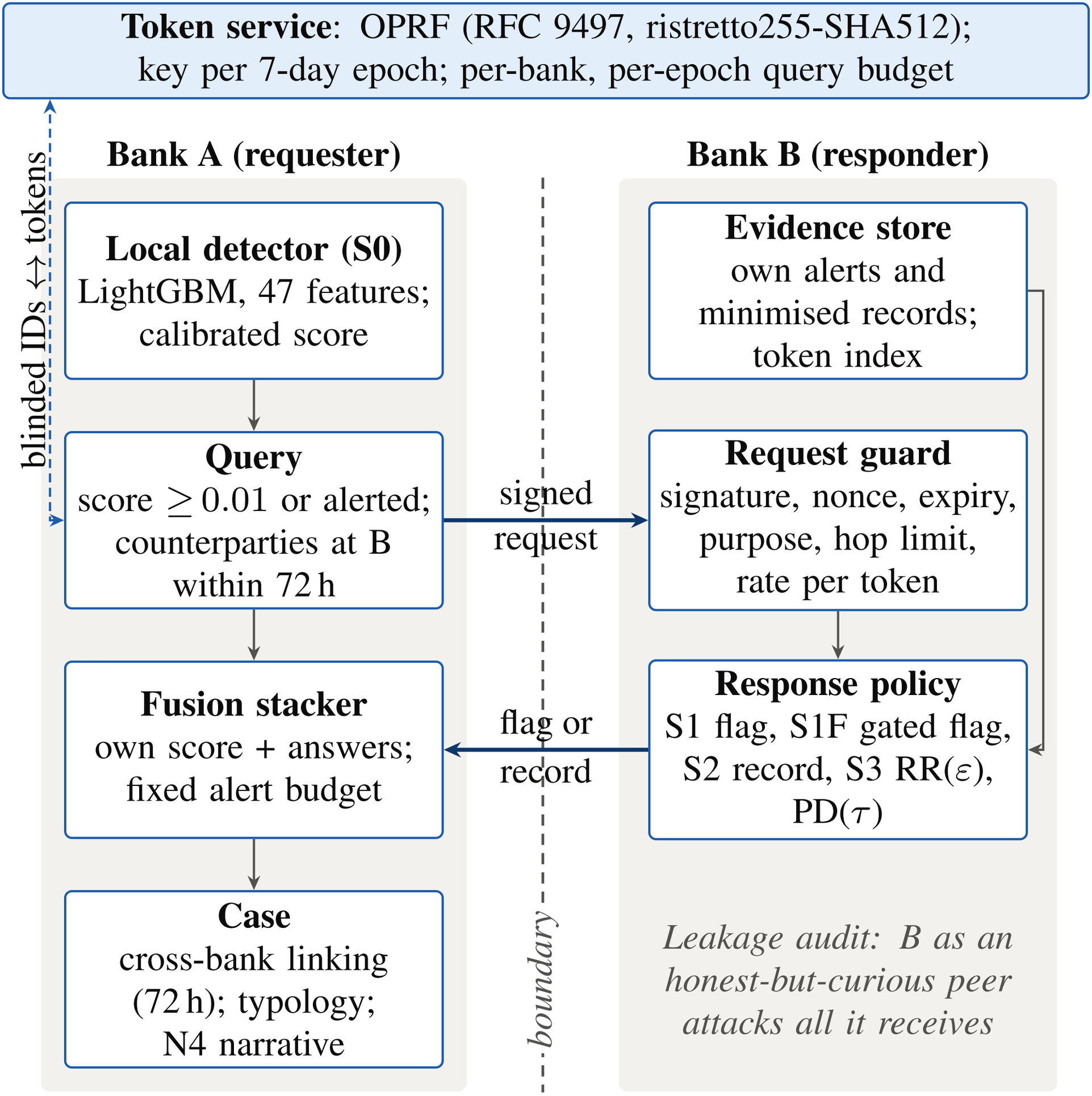

Token service: OPRF (RFC 9497, ristretto255-SHA512); key per 7-day epoch; per-bank, per-epoch query budget
blinded IDs ↔ tokens
Bank A (requester)
Bank B (responder)
Local detector (S0)
LightGBM, 47 features; calibrated score
Query
score ≥ 0.01 or alerted; counterparties at B within 72 h
Fusion stacker
own score + answers; fixed alert budget
Case
cross-bank linking (72 h); typology; N4 narrative
signed request
flag or record
boundary
Evidence store
own alerts and minimised records; token index
Request guard
signature, nonce, expiry, purpose, hop limit, rate per token
Response policy
S1 flag, S1F gated flag, S2 record, S3 RR(ε), PD(τ)
Leakage audit: B as an honest-but-curious peer attacks all it receives